

The gaming industry was an early adopter of motion tracking and gesture control. Today, the SDK helps PC developers learn how to use and integrate these new interface technologies in a simple and straightforward way. Developers will be able to develop applications in the world of education, business, 3D modeling, 3D printing, and more.

For more information on the SDK, please visit: intel.com/software/perceptual

NUANCE DRAGON ASSISTANT*: A CASE STUDY IN PERCEPTUAL COMPUTING

Voice-control technology is one aspect of perceptual computing that is already experiencing significant progress. Nuance, famous for its Dragon Naturally Speaking voice-recognition software, helped guide and facilitate implementation of the SDK in the speech and voice capacities for Dragon Assistant. Nuance Dragon Assistant software will integrate with Ultrabook devices to enable voice commands spanning several applications, including Media Player*, Facebook*, Twitter*, and Web browsers. This early implementation allows users to speak a command to the computer. For example, after a user says, "Search Amazon for lawn chairs," the Web browser opens, locates the Amazon.com site, and searches on the phrase "lawn chairs." Sharing that result or a specific URL with Facebook friends or Twitter followers is also a simple phrase away: "Share this page on Facebook."

Peter Mahoney, chief marketing officer at Nuance, explained to Intel® Software Adrenaline that, "Voice recognition is an extremely complex task—you must handle many different things, from audio, to voice models, to integrating with apps. We tried to make the tasks available at fairly high-level chunks to help make the developer's job of integration easier." With more power available, Nuance has greatly improved both the speed and accuracy of voice recognition by using multiple cores for better efficiency. That efficiency is becoming more important for Nuance and the Intel Perceptual Computing SDK as users increasingly expect longer battery life from their mobile devices.

As part of the SDK, Nuance's technology will be made available to developers. They can use it to add and expand



on the number of voice-controlled applications on Microsoft Windows* and Intel® platforms. Software developers can pass nearly all of the audio processing and handling to the development platform, providing Nuance an audio file and getting a basic text file result of the recognized terms. The application can then parse the result, using it in several ways. The goal is to make the use of voice-control technology simple and intuitive for developers to use.

As both Nuance's technology and the Intel Perceptual Computing SDK evolve, expect voice recognition to expand and become more refined, too. The next big push will come in the form of natural-language processing and its requirement for more processing horsepower and much larger, cloud-based databases. Natural-language analysis will allow computers to not only understand commands but also to understand a speaker's intent, an intensely more complex problem. Instead of being restricted to specific commands, natural language will allow users to, for example, say, "I need to find a new lawn chair," and then see a search result in their favorite shopping application.

MULTI-MODAL INTERACTION AND THE FUTURE OF PERCEPTUAL COMPUTING

The modalities of voice, facial recognition, and gesture are important. But the true power and potential for perceptual computing reside in the ability to combine these interfaces and sensors, and to increase the amount of data accessible to the computer (from the user); and to increase the amount of information the computer can receive at one time. Imagine a computer that can "see" if you are looking away from the screen while talking to determine whether it should interpret your voice as a command or as ambient noise.

A natural interface will allow an understanding of true intent, which can be acted upon in a fluid manner. Many visionaries believe that extensive perceptual computing is an attainable goal in the near future, and the Intel Percep-



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Open Innovation

Think outside the box. Think beyond the wires! Select this category for your idea that takes the features from the SDK and turns them into exciting, innovative experiences that free the user from traditional computing models. Establish your own rules and create the experience you once only imagined.